

Student Success Statistical Assistant

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Table of contents

1	Introduction	3
1.1	Industry	3
1.2	Intended User	3
1.3	Problem	3
1.4	Decision Supported	3
1.5	Project Goal	4
1.6	Selected Implementation Level	4
2	Dataset	4
2.1	Source	4
2.2	Main Variables	4
2.3	Data Import	5
2.4	Data Preparation and Quality	5
3	Explore Skill	6
3.1	User Question	6
3.2	Method	6
3.3	Visualizations	7
3.4	Zero-Grade Investigation	9
3.5	Validation	10
3.6	Interpretation	10
3.7	Limitation	10
4	Compare Skill	10
4.1	User Question	10
4.2	Variables	11
4.3	Method	11
4.4	Validation	13
4.5	Interpretation	13
4.6	Limitation	13

5	Model Skill	14
5.1	User Question	14
5.2	Outcome	14
5.3	Predictors	14
5.4	Model Type	14
5.5	Odds Ratios	16
5.6	Evaluation	17
5.7	Validation	18
5.8	Interpretation	19
5.9	Limitation	19
6	Additional Skill 1: Academic Progress Analysis	19
6.1	User Question	19
6.2	Method	19
6.3	Visualizations	20
6.4	Statistical Test	22
6.5	Validation	23
6.6	Interpretation	23
6.7	Limitation	23
7	Additional Skill 2: Program Support Prioritization	23
7.1	User Question	23
7.2	Method	23
7.3	Visualization	25
7.4	Validation	26
7.5	Interpretation	27
7.6	Limitation	27
8	Additional Skill 3: Data Quality Audit	27
8.1	User Question	27
8.2	Method	27
8.3	Readiness Result	30
8.4	Interpretation	30
8.5	Limitation	30
9	Validation Demonstrations	30
10	Google Antigravity and Agent Skills	31
10.1	Example 1: Explore Skill	31
10.2	Example 2: Compare Skill	32
10.3	Example 3: Model Skill	32
10.4	Statistical Verification	32
11	Application or Delivery	32

12 Limitations	33
12.1 Dataset Limitation	33
12.2 Statistical Limitation	33
12.3 Implementation Limitation	33
12.4 Additional Limitations	33
13 Reflection	34
14 Conclusion	34

1 Introduction

1.1 Industry

This project focuses on the higher-education industry.

1.2 Intended User

The intended user is a college student success coordinator or academic advisor.

1.3 Problem

Colleges need to understand patterns associated with student dropout and academic success so they can identify areas where students may benefit from earlier academic, financial, or advising support.

1.4 Decision Supported

The statistical assistant supports decisions involving:

- Academic advising
- Tutoring and academic support
- Financial-support review
- Program-level review
- Student-retention planning
- Data readiness and reliability

The results are intended to guide further investigation. They should not be used to automatically label students, deny services, or punish academic programs.

1.5 Project Goal

The goal is to create a reproducible statistical assistant that uses R and Agent Skills to examine student outcomes, validate requests, create useful results, and explain findings in plain language.

1.6 Selected Implementation Level

This project uses **Level 1: Quarto Statistical Report**.

Level 1 was selected because the project focuses on R, statistics, reproducibility, interpretation, validation, and Agent Skills. A separate application programming interface or interactive application is not required.

2 Dataset

2.1 Source

The project uses the UCI Machine Learning Repository dataset titled **Predict Students' Dropout and Academic Success**.

One row represents one college student.

The dataset contains:

- 4,424 observations
- 37 variables
- 1,421 Dropout outcomes
- 794 Enrolled outcomes
- 2,209 Graduate outcomes

2.2 Main Variables

Important variables include:

- target
- course
- tuition_fees_up_to_date
- debtor
- scholarship_holder
- admission_grade
- age_at_enrollment

- curricular_units_1st_sem_enrolled
- curricular_units_1st_sem_approved
- curricular_units_1st_sem_grade

2.3 Data Import

```
library(tidyverse)
library(janitor)

students <- read_delim(
  "data.csv",
  delim = ";",
  show_col_types = FALSE
) |>
  clean_names()

dim(students)
```

```
[1] 4424  37
```

```
table(students$target)
```

```
Dropout Enrolled Graduate
      1421       794      2209
```

2.4 Data Preparation and Quality

The variable names were standardized using `clean_names()`.

The dataset was checked for missing values, duplicate rows, invalid categories, negative academic values, and logical inconsistencies.

```
missing_values <- colSums(is.na(students))
duplicate_rows <- sum(duplicated(students))

sum(missing_values)
```

```
[1] 0
```

```
duplicate_rows
```

```
[1] 0
```

The dataset contained zero missing values and zero completely duplicated rows.

Unusual zero values were investigated rather than automatically removed.

3 Explore Skill

3.1 User Question

What does the distribution of first-semester grades look like?

3.2 Method

The Explore skill uses descriptive statistics, a histogram, a boxplot, and an investigation of zero-grade records.

The selected variable is:

curricular_units_1st_sem_grade

```
explore_summary <- students |>
  summarise(
    observations = n(),
    missing_values =
      sum(is.na(curricular_units_1st_sem_grade)),
    mean_grade =
      mean(curricular_units_1st_sem_grade, na.rm = TRUE),
    median_grade =
      median(curricular_units_1st_sem_grade, na.rm = TRUE),
    standard_deviation =
      sd(curricular_units_1st_sem_grade, na.rm = TRUE),
    minimum =
      min(curricular_units_1st_sem_grade, na.rm = TRUE),
    first_quartile =
      quantile(
        curricular_units_1st_sem_grade,
        0.25,
```

```

    na.rm = TRUE
  ),
  third_quartile =
    quantile(
      curricular_units_1st_sem_grade,
      0.75,
      na.rm = TRUE
    ),
  maximum =
    max(curricular_units_1st_sem_grade, na.rm = TRUE)
)

explore_summary

```

```

# A tibble: 1 x 9
  observations missing_values mean_grade median_grade standard_deviation minimum
      <int>         <int>      <dbl>      <dbl>          <dbl>      <dbl>
1         4424           0      10.6       12.3           4.84         0
# i 3 more variables: first_quartile <dbl>, third_quartile <dbl>, maximum <dbl>

```

The results showed:

- Mean: approximately 10.64
- Median: approximately 12.29
- Standard deviation: approximately 4.84
- Minimum: 0
- First quartile: 11
- Third quartile: 13.40
- Maximum: 18.88

3.3 Visualizations

```

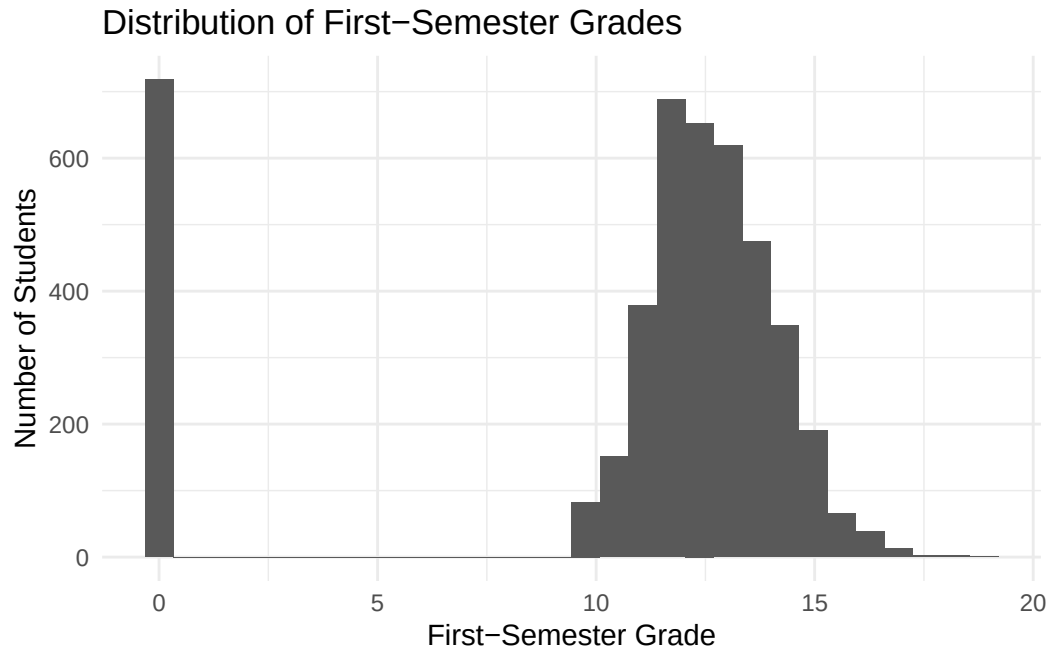
ggplot(
  students,
  aes(x = curricular_units_1st_sem_grade)
) +
  geom_histogram(bins = 30) +
  labs(
    title = "Distribution of First-Semester Grades",
    x = "First-Semester Grade",

```

```

  y = "Number of Students"
) +
  theme_minimal()

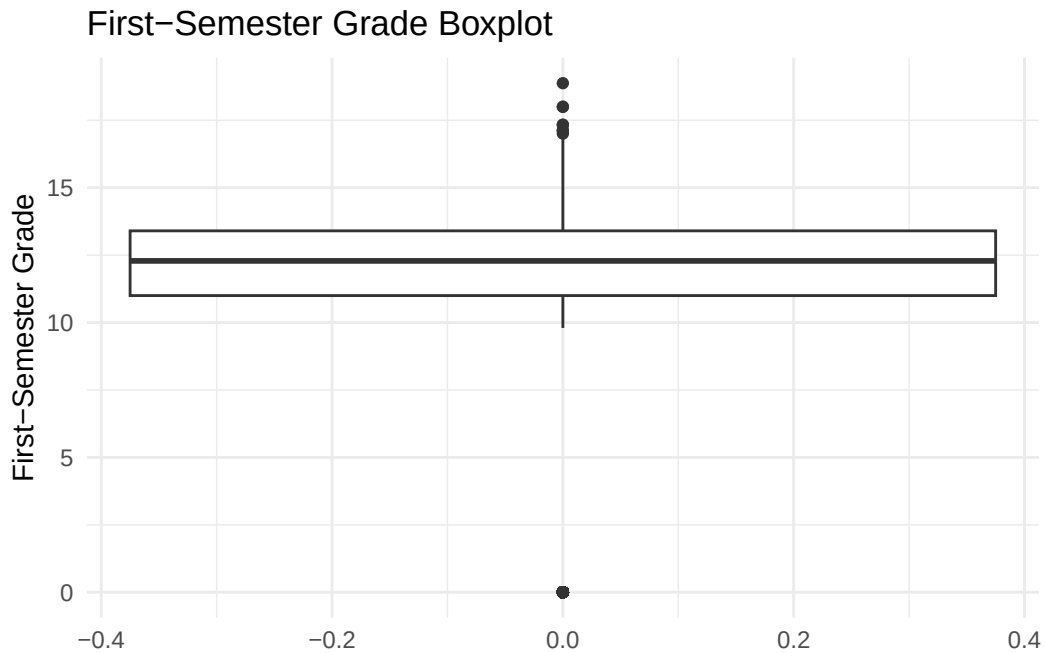
```



```

ggplot(
  students,
  aes(y = curricular_units_1st_sem_grade)
) +
  geom_boxplot() +
  labs(
    title = "First-Semester Grade Boxplot",
    y = "First-Semester Grade"
  ) +
  theme_minimal()

```

3.4 Zero-Grade Investigation

```
zero_grade_summary <- students |>
  filter(curricular_units_1st_sem_grade == 0) |>
  summarise(
    zero_grade_count = n(),
    zero_grade_percentage =
      n() / nrow(students) * 100,
    average_units_enrolled =
      mean(curricular_units_1st_sem_enrolled),
    average_units_approved =
      mean(curricular_units_1st_sem_approved),
    average_without_evaluations =
      mean(curricular_units_1st_sem_without_evaluations)
  )

zero_grade_summary
```

```
# A tibble: 1 x 5
  zero_grade_count zero_grade_percentage average_units_enrolled
      <int>             <dbl>             <dbl>
```

```
1           718           16.2           4.16
# i 2 more variables: average_units_approved <dbl>,
#   average_without_evaluations <dbl>
```

There were 718 zero-grade records, representing approximately 16.23% of the dataset.

All 718 students with a zero first-semester grade approved zero first-semester units. These values appear to contain meaningful academic information and were not removed.

3.5 Validation

The skill confirms that the selected variable exists and is numeric before calculating numeric summaries.

It also counts missing values and investigates unusual values instead of silently deleting them.

3.6 Interpretation

The grade distribution contains many zero values, which pull the mean below the median.

Most nonzero grades are concentrated around the middle and upper part of the observed range.

The results describe the distribution but do not explain why students received particular grades.

3.7 Limitation

This is a descriptive analysis. It does not establish that first-semester grades caused later student outcomes.

4 Compare Skill

4.1 User Question

Does the dropout percentage differ according to whether tuition fees are up to date?

4.2 Variables

The grouping variable is:

tuition_fees_up_to_date

The outcome was prepared as:

- Dropout
- Not Dropout

Enrolled and Graduate outcomes were combined into the Not Dropout category.

4.3 Method

The analysis calculated group counts, dropout percentages, the difference in proportions, a 95% confidence interval, and a two-sample proportion test.

```
compare_data <- students |>
  mutate(
    dropout_status = if_else(
      target == "Dropout",
      "Dropout",
      "Not Dropout"
    ),
    tuition_status = if_else(
      tuition_fees_up_to_date == 1,
      "Up to date",
      "Not up to date"
    )
  )

compare_summary <- compare_data |>
  group_by(tuition_status) |>
  summarise(
    sample_size = n(),
    dropout_count =
      sum(dropout_status == "Dropout"),
    not_dropout_count =
      sum(dropout_status == "Not Dropout"),
    dropout_percentage =
      mean(dropout_status == "Dropout") * 100,
    .groups = "drop"
```

```
)

compare_summary
```

```
# A tibble: 2 x 5
  tuition_status sample_size dropout_count not_dropout_count dropout_percentage
  <chr>           <int>         <int>         <int>         <dbl>
1 Not up to date     528           457           71           86.6
2 Up to date        3896           964          2932           24.7
```

Students whose tuition was not up to date had an observed dropout percentage of approximately 86.55%.

Students whose tuition was up to date had an observed dropout percentage of approximately 24.74%.

```
dropout_counts <- c(
  sum(
    compare_data$dropout_status == "Dropout" &
    compare_data$tuition_status == "Not up to date"
  ),
  sum(
    compare_data$dropout_status == "Dropout" &
    compare_data$tuition_status == "Up to date"
  )
)

group_sizes <- c(
  sum(compare_data$tuition_status == "Not up to date"),
  sum(compare_data$tuition_status == "Up to date")
)

comparison_test <- prop.test(
  x = dropout_counts,
  n = group_sizes,
  correct = FALSE
)

comparison_test
```

2-sample test for equality of proportions without continuity correction

```
data: dropout_counts out of group_sizes
X-squared = 814.76, df = 1, p-value < 2.2e-16
alternative hypothesis: two.sided
95 percent confidence interval:
 0.5859975 0.6501966
sample estimates:
   prop 1    prop 2 
0.8655303 0.2474333
```

The estimated difference was approximately 61.81 percentage points.

The 95% confidence interval for the difference was approximately 58.60 to 65.02 percentage points.

The proportion test produced a p-value below 2.2×10^{-16} .

4.4 Validation

The grouping variable must contain exactly two valid groups.

Both groups must contain enough observations, and the outcome values must be valid before the test is performed.

4.5 Interpretation

There was a large and statistically significant association between tuition-payment status and dropout status.

Students whose tuition fees were not up to date had a substantially higher observed dropout percentage.

This result may support further investigation of financial barriers and student-support options.

4.6 Limitation

This observational analysis identifies an association. It does not prove that unpaid tuition caused students to drop out.

Combining Enrolled and Graduate students into one Not Dropout group also simplifies the original three-category outcome.

5 Model Skill

5.1 User Question

Which factors are associated with whether a student drops out or graduates?

5.2 Outcome

The binary model outcome was coded as:

- Graduate = 0
- Dropout = 1

Students whose outcome was Enrolled were excluded from this model.

5.3 Predictors

The model included:

- First-semester grade
- First-semester approved units
- Tuition-fee status
- Debtor status
- Scholarship-holder status
- Admission grade
- Age at enrollment

5.4 Model Type

A binary logistic regression model was used.

```
model_data <- students |>
  filter(target %in% c("Dropout", "Graduate")) |>
  mutate(
    dropout_outcome =
      if_else(target == "Dropout", 1, 0),
    tuition_fees_up_to_date =
      factor(tuition_fees_up_to_date),
    debtor = factor(debtor),
    scholarship_holder =
      factor(scholarship_holder)
```

```

) |>
select(
  dropout_outcome,
  curricular_units_1st_sem_grade,
  curricular_units_1st_sem_approved,
  tuition_fees_up_to_date,
  debtor,
  scholarship_holder,
  admission_grade,
  age_at_enrollment
) |>
drop_na()

set.seed(123)

training_rows <- sample(
  seq_len(nrow(model_data)),
  size = floor(0.80 * nrow(model_data))
)

training_data <- model_data[training_rows, ]
test_data <- model_data[-training_rows, ]

dropout_model <- glm(
  dropout_outcome ~
    curricular_units_1st_sem_grade +
    curricular_units_1st_sem_approved +
    tuition_fees_up_to_date +
    debtor +
    scholarship_holder +
    admission_grade +
    age_at_enrollment,
  data = training_data,
  family = binomial()
)

summary(dropout_model)

```

Call:

```

glm(formula = dropout_outcome ~ curricular_units_1st_sem_grade +
    curricular_units_1st_sem_approved + tuition_fees_up_to_date +

```

```
debtor + scholarship_holder + admission_grade + age_at_enrollment,
family = binomial(), data = training_data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	5.950099	0.597040	9.966	< 2e-16 ***
curricular_units_1st_sem_grade	-0.066509	0.018536	-3.588	0.000333 ***
curricular_units_1st_sem_approved	-0.410190	0.032987	-12.435	< 2e-16 ***
tuition_fees_up_to_date1	-2.970707	0.266024	-11.167	< 2e-16 ***
debtor1	1.024425	0.200231	5.116	3.12e-07 ***
scholarship_holder1	-1.289320	0.143259	-9.000	< 2e-16 ***
admission_grade	-0.016888	0.003733	-4.524	6.07e-06 ***
age_at_enrollment	0.051412	0.006872	7.482	7.33e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 3895.9 on 2903 degrees of freedom
Residual deviance: 2174.8 on 2896 degrees of freedom
AIC: 2190.8

Number of Fisher Scoring iterations: 6

5.5 Odds Ratios

```
odds_ratios <- exp(coef(dropout_model))
odds_ratios
```

(Intercept)	curricular_units_1st_sem_grade
383.79128313	0.93565437
curricular_units_1st_sem_approved	tuition_fees_up_to_date1
0.66352386	0.05126703
debtor1	scholarship_holder1
2.78549285	0.27545816
admission_grade	age_at_enrollment
0.98325384	1.05275670

The model results indicated:

- Higher first-semester grades were associated with lower odds of dropout.

- More approved first-semester units were associated with lower odds of dropout.
- Having tuition fees up to date was associated with substantially lower odds of dropout.
- Debtor status was associated with higher odds of dropout.
- Scholarship-holder status was associated with lower odds of dropout.
- Higher admission grades were associated with slightly lower odds of dropout.
- Greater age at enrollment was associated with higher estimated odds of dropout.

5.6 Evaluation

```

predicted_probability <- predict(
  dropout_model,
  newdata = test_data,
  type = "response"
)

predicted_class <- if_else(
  predicted_probability >= 0.50,
  1,
  0
)

confusion_matrix <- table(
  Predicted = predicted_class,
  Actual = test_data$dropout_outcome
)

confusion_matrix

```

	Actual	
Predicted	0	1
0	421	73
1	30	202

```

true_negative <- confusion_matrix["0", "0"]
false_negative <- confusion_matrix["0", "1"]
false_positive <- confusion_matrix["1", "0"]
true_positive <- confusion_matrix["1", "1"]

accuracy <- (
  true_positive + true_negative

```

```

) / sum(confusion_matrix)

sensitivity <- true_positive / (
  true_positive + false_negative
)

specificity <- true_negative / (
  true_negative + false_positive
)

model_metrics <- tibble(
  accuracy = accuracy,
  sensitivity = sensitivity,
  specificity = specificity
)

model_metrics

```

```

# A tibble: 1 x 3
  accuracy sensitivity specificity
  <dbl>         <dbl>         <dbl>
1   0.858         0.735         0.933

```

The test-set results were approximately:

- Accuracy: 85.81%
- Sensitivity: 73.45%
- Specificity: 93.35%

The model was better at identifying graduates than dropouts.

5.7 Validation

The model skill confirms that:

- The outcome contains exactly two categories.
- All required predictors exist.
- Required values are available.
- The selected model is appropriate for a binary outcome.
- Prediction requests contain all required inputs.

5.8 Interpretation

The model identifies broad factors associated with Dropout versus Graduate outcomes.

It may support early review and outreach, but it should not be used to make automatic decisions about individual students.

5.9 Limitation

The model excludes Enrolled students and may make classification errors.

The results are associative and do not prove that any predictor caused dropout.

6 Additional Skill 1: Academic Progress Analysis

6.1 User Question

How does first-semester academic progress differ among students who dropped out, remained enrolled, or graduated?

6.2 Method

An approval-rate variable was created using approved units divided by enrolled units.

Rates were not calculated when enrolled units equaled zero.

```
academic_progress <- students |>
  mutate(
    approval_rate = if_else(
      curricular_units_1st_sem_enrolled > 0,
      curricular_units_1st_sem_approved /
        curricular_units_1st_sem_enrolled,
      NA_real_
    )
  )

academic_summary <- academic_progress |>
  group_by(target) |>
  summarise(
    sample_size = n(),
    mean_grade =
```

```

    mean(curricular_units_1st_sem_grade, na.rm = TRUE),
  median_grade =
    median(curricular_units_1st_sem_grade, na.rm = TRUE),
  mean_units_enrolled =
    mean(curricular_units_1st_sem_enrolled, na.rm = TRUE),
  mean_units_approved =
    mean(curricular_units_1st_sem_approved, na.rm = TRUE),
  mean_approval_rate =
    mean(approval_rate, na.rm = TRUE) * 100,
  median_approval_rate =
    median(approval_rate, na.rm = TRUE) * 100,
  .groups = "drop"
)

academic_summary

```

```

# A tibble: 3 x 8
  target      sample_size mean_grade median_grade mean_units_enrolled
  <chr>          <int>      <dbl>      <dbl>          <dbl>
1 Dropout         1421        7.26        10.9            5.82
2 Enrolled          794        11.1         12             5.96
3 Graduate        2209        12.6         13             6.67
# i 3 more variables: mean_units_approved <dbl>, mean_approval_rate <dbl>,
#   median_approval_rate <dbl>

```

The average approval rates were approximately:

- Dropout: 40.60%
- Enrolled: 71.35%
- Graduate: 93.50%

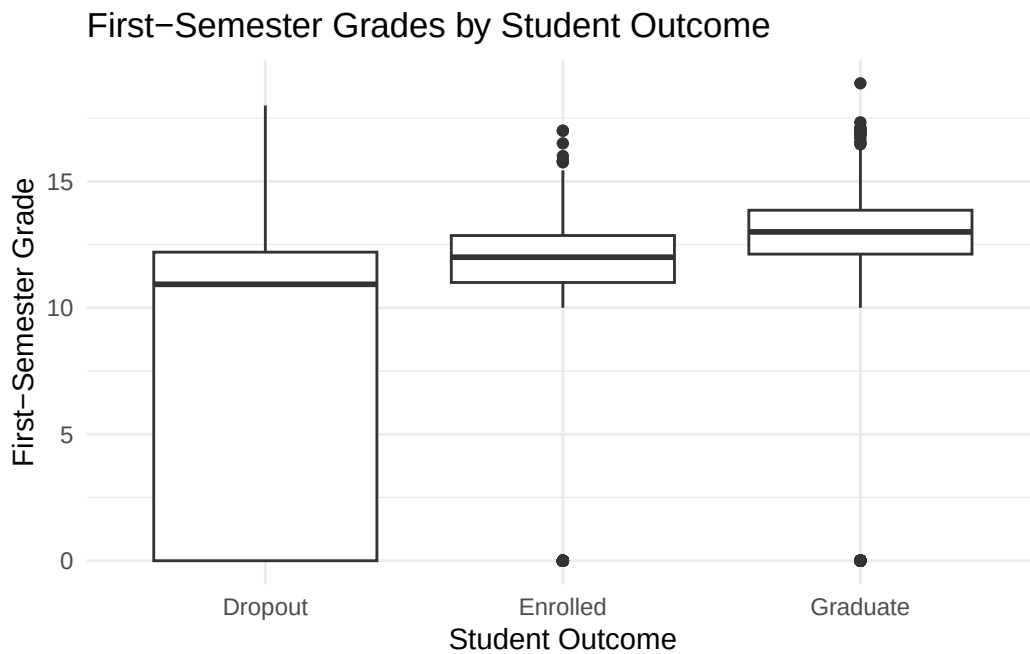
6.3 Visualizations

```

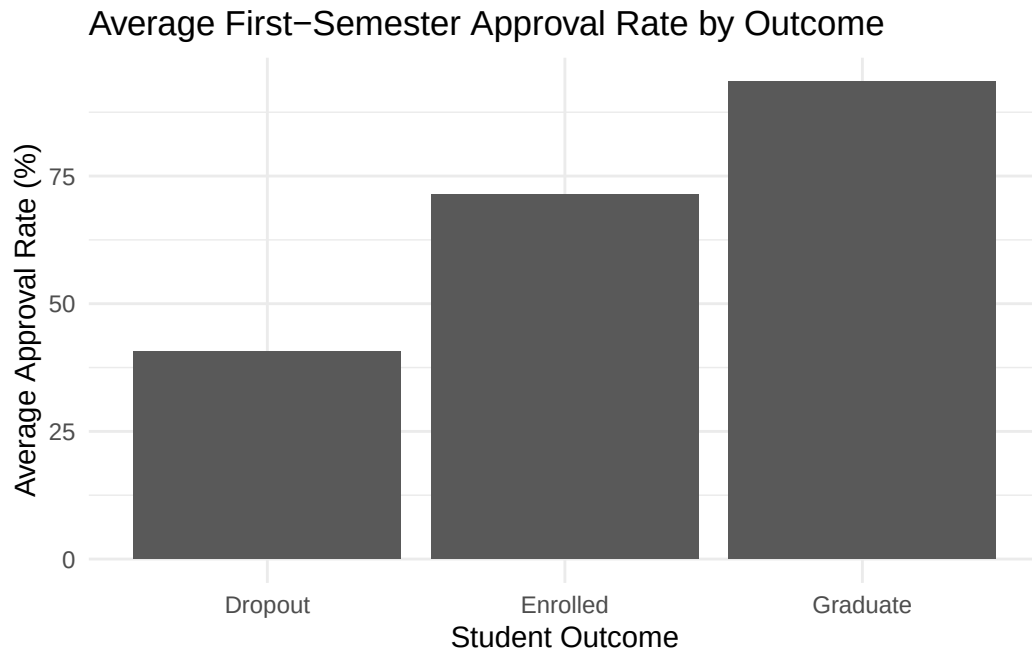
ggplot(
  academic_progress,
  aes(
    x = target,
    y = curricular_units_1st_sem_grade
  )
) +

```

```
geom_boxplot() +
labs(
  title = "First-Semester Grades by Student Outcome",
  x = "Student Outcome",
  y = "First-Semester Grade"
) +
theme_minimal()
```



```
ggplot(
  academic_summary,
  aes(
    x = target,
    y = mean_approval_rate
  )
) +
geom_col() +
labs(
  title = "Average First-Semester Approval Rate by Outcome",
  x = "Student Outcome",
  y = "Average Approval Rate (%)"
) +
theme_minimal()
```



6.4 Statistical Test

```
grade_test <- kruskal.test(  
  curricular_units_1st_sem_grade ~ target,  
  data = academic_progress  
)  
  
grade_test
```

Kruskal-Wallis rank sum test

data: curricular_units_1st_sem_grade by target
Kruskal-Wallis chi-squared = 1094.2, df = 2, p-value < 2.2e-16

The Kruskal-Wallis test produced:

- Chi-squared statistic: 1094.2
- Degrees of freedom: 2
- P-value: below 2.2×10^{-16}

6.5 Validation

The analysis found:

- Zero negative enrolled-unit values
- Zero negative approved-unit values
- Zero cases where approved units exceeded enrolled units
- Zero approval rates outside the range of 0 to 1
- 180 missing approval rates caused by zero enrolled units

6.6 Interpretation

Graduate students showed the strongest first-semester academic progress.

Enrolled students were in the middle, while Dropout students had the lowest average grades, approved units, and approval rates.

6.7 Limitation

This retrospective analysis does not prove that first-semester academic performance caused later outcomes.

7 Additional Skill 2: Program Support Prioritization

7.1 User Question

Which academic programs have the highest observed dropout percentages and may require further review?

7.2 Method

The analysis calculated program sample sizes, outcome counts, outcome percentages, sample-size warnings, rankings, and approximate 95% confidence intervals.

```
program_summary <- students |>
  group_by(course) |>
  summarise(
    sample_size = n(),
    dropout_count =
      sum(target == "Dropout"),
```

```

    enrolled_count =
      sum(target == "Enrolled"),
    graduate_count =
      sum(target == "Graduate"),
    dropout_percentage =
      mean(target == "Dropout") * 100,
    enrolled_percentage =
      mean(target == "Enrolled") * 100,
    graduate_percentage =
      mean(target == "Graduate") * 100,
    .groups = "drop"
  ) |>
  arrange(desc(dropout_percentage)) |>
  mutate(
    sample_size_warning = if_else(
      sample_size < 30,
      "Small sample - interpret carefully",
      "Adequate sample size"
    ),
    dropout_proportion =
      dropout_count / sample_size,
    standard_error =
      sqrt(
        dropout_proportion *
          (1 - dropout_proportion) /
          sample_size
      ),
    confidence_low =
      pmax(
        0,
        dropout_proportion -
          1.96 * standard_error
      ) * 100,
    confidence_high =
      pmin(
        1,
        dropout_proportion +
          1.96 * standard_error
      ) * 100
  )

program_summary |>

```



```
slice_head(n = 5)
```

```
# A tibble: 5 x 13
  course sample_size dropout_count enrolled_count graduate_count
  <dbl>      <int>      <int>      <int>      <int>
1     33         12          8          3          1
2    9130        141         78         21         42
3    9119        170         92         64         14
4    9991        268        136         54         78
5   9853        192         85         50         57
# i 8 more variables: dropout_percentage <dbl>, enrolled_percentage <dbl>,
# graduate_percentage <dbl>, sample_size_warning <chr>,
# dropout_proportion <dbl>, standard_error <dbl>, confidence_low <dbl>,
# confidence_high <dbl>
```

Course 33 had the highest observed dropout percentage, but it contained only 12 students and was flagged as a small sample.

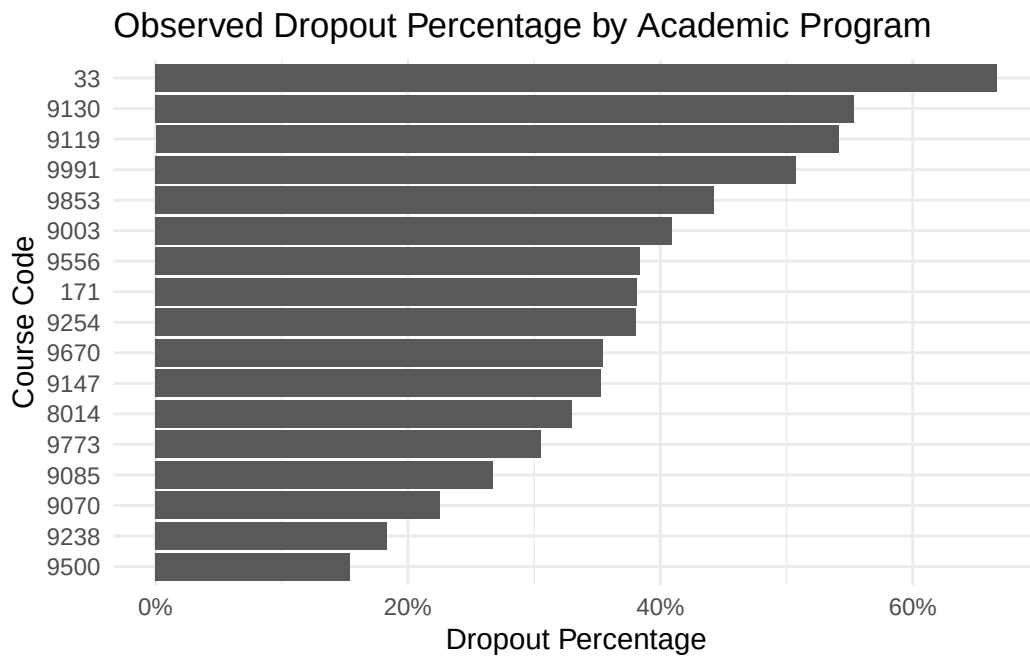
Among programs with adequate sample sizes, the highest observed dropout percentages occurred in:

- Course 9130
- Course 9119
- Course 9991
- Course 9853

7.3 Visualization

```
ggplot(
  program_summary,
  aes(
    x = reorder(
      as.factor(course),
      dropout_percentage
    ),
    y = dropout_percentage
  )
) +
  geom_col() +
  coord_flip() +
```

```
labs(
  title = "Observed Dropout Percentage by Academic Program",
  x = "Course Code",
  y = "Dropout Percentage"
) +
scale_y_continuous(
  labels = scales::label_percent(scale = 1)
) +
theme_minimal()
```



7.4 Validation

The dataset contained:

- Zero missing course values
- Zero missing target values
- Zero unexpected target categories
- 17 academic programs

Programs with fewer than 30 students were flagged.

7.5 Interpretation

The results help administrators identify programs that may deserve further investigation.

A high dropout percentage should be considered together with sample size and confidence intervals.

7.6 Limitation

A high program dropout percentage does not prove that the program caused students to leave or that the program is low quality.

8 Additional Skill 3: Data Quality Audit

8.1 User Question

Are missing values, duplicates, invalid codes, or logical inconsistencies affecting the reliability of the dataset?

8.2 Method

The audit examined missing values, duplicates, invalid categories, negative values, and academic logic.

```
data_quality_summary <- tibble(  
  observations = nrow(students),  
  variables = ncol(students),  
  total_missing_values =  
    sum(is.na(students)),  
  duplicate_rows =  
    sum(duplicated(students))  
)
```

```
data_quality_summary
```

```
# A tibble: 1 x 4
```

	observations	variables	total_missing_values	duplicate_rows
	<int>	<int>	<int>	<int>
1	4424	37	0	0

```
category_checks <- tibble(
  invalid_target =
    sum(
      !students$target %in%
        c("Dropout", "Enrolled", "Graduate")
    ),
  invalid_tuition_status =
    sum(
      !students$tuition_fees_up_to_date %in%
        c(0, 1)
    ),
  invalid_debtor_status =
    sum(
      !students$debtor %in%
        c(0, 1)
    ),
  invalid_scholarship_status =
    sum(
      !students$scholarship_holder %in%
        c(0, 1)
    )
)

category_checks
```

```
# A tibble: 1 x 4
  invalid_target invalid_tuition_status invalid_debtor_status
      <int>             <int>             <int>
1         0               0               0
# i 1 more variable: invalid_scholarship_status <int>
```

```
academic_checks <- tibble(
  negative_enrolled =
    sum(
      students$curricular_units_1st_sem_enrolled < 0
    ),
  negative_approved =
    sum(
      students$curricular_units_1st_sem_approved < 0
    ),
  approved_greater_than_enrolled =
```

```

sum(
  students$curricular_units_1st_sem_approved >
    students$curricular_units_1st_sem_enrolled
),
negative_grade =
sum(
  students$curricular_units_1st_sem_grade < 0
),
zero_first_semester_grade =
sum(
  students$curricular_units_1st_sem_grade == 0
),
zero_enrolled_units =
sum(
  students$curricular_units_1st_sem_enrolled == 0
)
)

academic_checks

```

```

# A tibble: 1 x 6
  negative_enrolled negative_approved approved_greater_than_enr~1 negative_grade
      <int>          <int>          <int>          <int>
1         0            0            0            0
# i abbreviated name: 1: approved_greater_than_enrolled
# i 2 more variables: zero_first_semester_grade <int>,
#   zero_enrolled_units <int>

```

The audit found:

- 0 missing values
- 0 duplicate rows
- 0 invalid target values
- 0 invalid financial-status codes
- 0 negative academic values
- 0 cases where approved units exceeded enrolled units
- 718 zero first-semester grades
- 180 students with zero enrolled units

8.3 Readiness Result

```
records_affected <- 0
records_affected_percentage <- 0
data_quality_score <- 100
readiness_result <- "Ready for analysis"

quality_result <- tibble(
  records_affected,
  records_affected_percentage,
  data_quality_score,
  readiness_result
)

quality_result
```

```
# A tibble: 1 x 4
  records_affected records_affected_percen~1 data_quality_score readiness_result
      <dbl>             <dbl>             <dbl> <chr>
1           0              0              100 Ready for analy~
# i abbreviated name: 1: records_affected_percentage
```

8.4 Interpretation

The dataset passed all documented automated quality checks and was classified as ready for analysis.

The zero values were retained because unusual values are not automatically errors.

8.5 Limitation

A data-quality score of 100 does not prove that every value is correct.

Some incorrect values may still fall within expected ranges.

9 Validation Demonstrations

The assistant should stop when a request is invalid or inappropriate.

The project demonstrates the following invalid requests:

1. A selected variable does not exist.
2. A numeric analysis is requested for a categorical variable.
3. A grouping variable contains three groups instead of exactly two.
4. A group contains fewer than 30 observations.
5. A causal conclusion is requested from observational data.
6. A prediction is requested without all required inputs.

Example messages include:

- The selected variable does not exist in the dataset.
- The selected variable must be numeric for this analysis.
- The grouping variable must contain exactly two valid groups.
- At least one group contains fewer than 30 observations.
- This analysis identifies an association. It does not prove causation.
- Prediction cannot be completed because required inputs are missing.

These checks reduce the risk of producing incorrect or misleading results.

10 Google Antigravity and Agent Skills

Google Antigravity was used to help create, organize, and review the Agent Skills.

10.1 Example 1: Explore Skill

Antigravity was asked to create a skill for exploring a numeric student variable.

The file created was:

```
.agents/skills/explore/SKILL.md
```

The student reviewed the descriptive statistics, validation steps, visualizations, and interpretation instructions.

The recommendation to investigate zero values rather than automatically remove them was accepted.

10.2 Example 2: Compare Skill

Antigravity was asked to create a skill comparing dropout percentages between two tuition-status groups.

The file created was:

```
.agents/skills/compare/SKILL.md
```

The student reviewed the two-group requirement, confidence interval, proportion test, and interpretation.

The student ensured that the final language described an association rather than causation.

10.3 Example 3: Model Skill

Antigravity was asked to create a logistic-regression skill for Dropout versus Graduate outcomes.

The file created was:

```
.agents/skills/model/SKILL.md
```

The student reviewed the predictors, train-test split, odds ratios, and evaluation metrics.

A simpler base-R evaluation workflow was used after additional modeling packages caused installation problems.

10.4 Statistical Verification

All statistical results were verified by running the R code in Google Colab.

The student reviewed the generated tables, model output, validation results, and visualizations before accepting the interpretations.

11 Application or Delivery

The intended user accesses the assistant through the Quarto statistical report.

The user can review:

- Statistical questions
- Methods
- Tables
- Visualizations

- Validation messages
- Practical interpretations
- Limitations

The HTML version can be opened in a browser, while the PDF can be used for submission and offline viewing.

12 Limitations

12.1 Dataset Limitation

The dataset represents a particular educational context and may not generalize to every college, country, or student population.

12.2 Statistical Limitation

The analyses use observational data and identify associations. They do not prove causation.

The logistic regression model also excludes students whose outcome was Enrolled.

12.3 Implementation Limitation

The Level 1 implementation is a static Quarto report.

It does not provide a live interface, API, or automatic analysis of newly submitted student records.

12.4 Additional Limitations

- Small academic programs may have unstable percentages.
- Some relevant student circumstances may not be included in the dataset.
- Automated validation cannot detect every possible data error.
- Model predictions may contain false positives and false negatives.

13 Reflection

This project helped me learn how to organize a complete statistical analysis using R, Quarto, GitHub, and Agent Skills.

One difficult part was organizing the code into separate files and connecting each statistical question to the appropriate method.

Another difficult part was deciding how to handle unusual zero values. The zero-grade records could not simply be removed because they appeared to contain meaningful information.

Building and interpreting the logistic regression model was also challenging because the coefficients and odds ratios needed to be explained in plain language.

The three additional skills were selected because they support different parts of the higher-education problem:

- Academic Progress Analysis examines early academic performance.
- Program Support Prioritization supports program-level review.
- Data Quality Audit confirms whether the dataset is ready for analysis.

The original Financial Risk skill idea was replaced because it overlapped with the required tuition-status comparison.

With more time, I would:

- Add readable names for every academic program.
- Add pairwise tests after the Kruskal-Wallis test.
- Test additional model types.
- Add cross-validation.
- Create an interactive Quarto website or application.
- Test the assistant with another higher-education dataset.

14 Conclusion

The Student Success Statistical Assistant combines six R-based skills into one reproducible report.

The project provides descriptive analysis, group comparison, logistic regression, academic-progress analysis, program prioritization, and data-quality validation.

The results may help college staff identify patterns that deserve further investigation while emphasizing responsible interpretation, validation, and limitations.